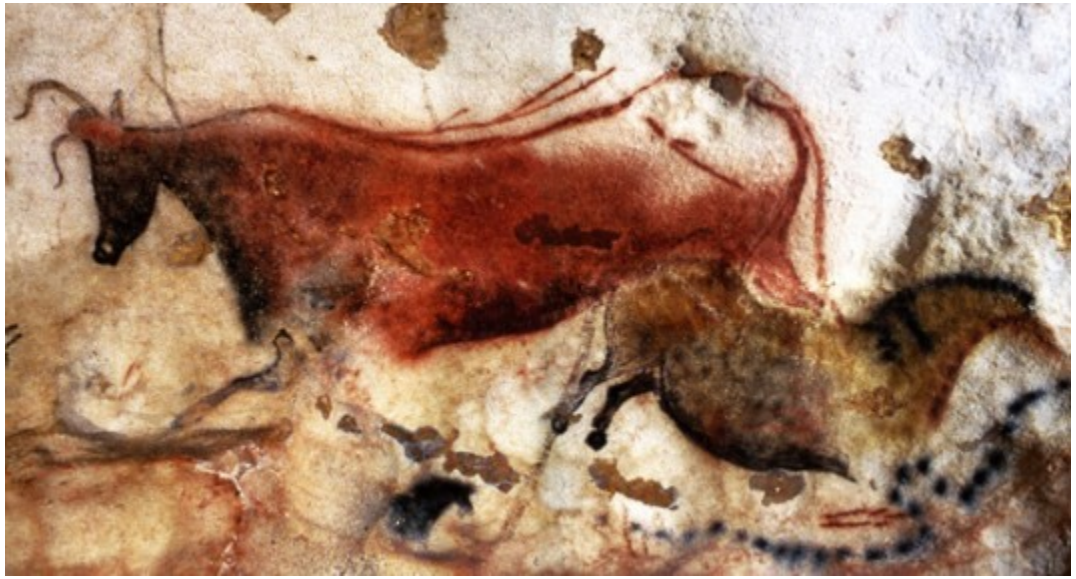
Hominid Evolutionary Trends



Cultural Evolution

Intro

- In addition to anatomical and physiological evolution, hominins have developed complex cultures that allow them to survive better in their environment.

- Cultural evolution can be seen in:
  - Tool and technology development
  - Development of methods for obtaining food
  - Increasing sophistication of language
  - Other changes including symbolic representation (eg visual art)

Australopithecine Culture

- Home bases:
  - Lived in one place, left from there to forage for food
- Use of fire?
  - No evidence found that Australopithecines deliberately used or controlled fire.
- Tools
  - Limited evidence that tools were made, but rocks and sticks were likely used as tools.



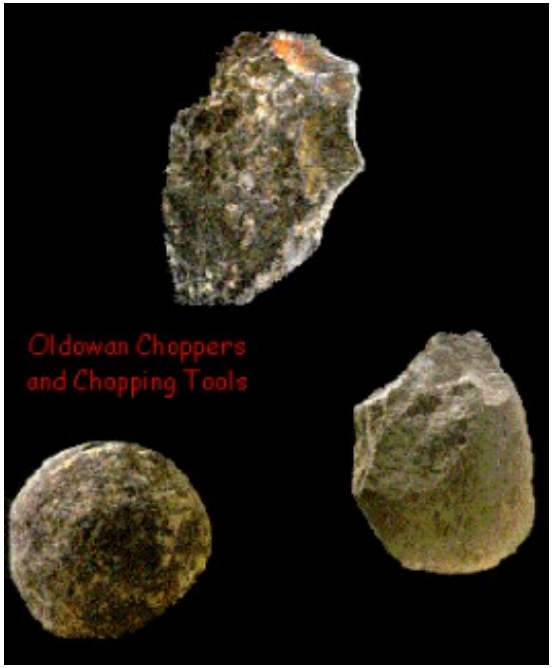
Early homo

- Larger brains than Australopithecines
- Needed more complex fats and higher levels of protein
- Therefore needed to consume meat.
- Food sharing from large hunts would have been necessary – collaboration and interdependence
- Males likely hunted, females gathered
- Majority of food from gathering, with occasional large bounty from hunt
- Cooperative culture requires development of language – not as complex as modern humans
- Evidence of hunting: stone tools and also marks on bone to indicate tools not teeth used.

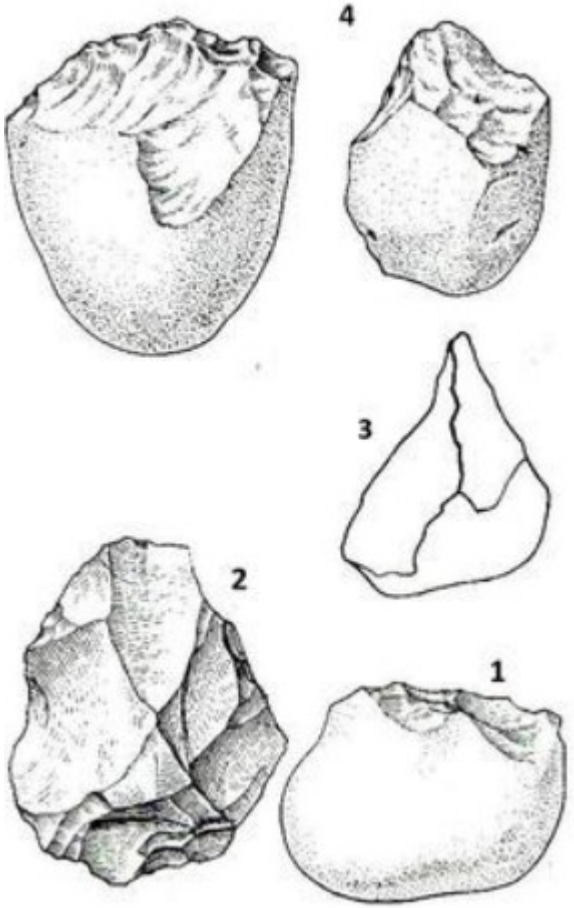
Homo habilis culture

- Tool Use – Oldowan Tools
  - Range of pebble tools found: choppers, scrapers, flakers, chisels
  - Require precision grip
  - Tool use enabled colonisation of a broader range of habitats





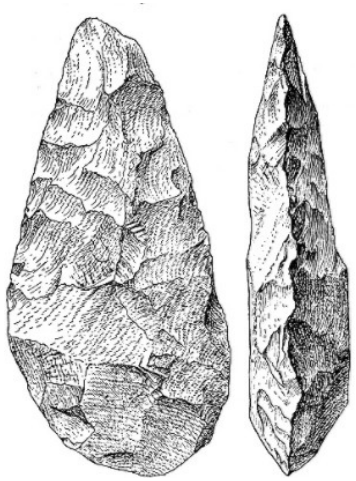
Oldowan tools examples



Note

- Rough flaking
- Unsophisticated
- Not developed for specific purpose eg arrowhead/fish hook/knife etc.

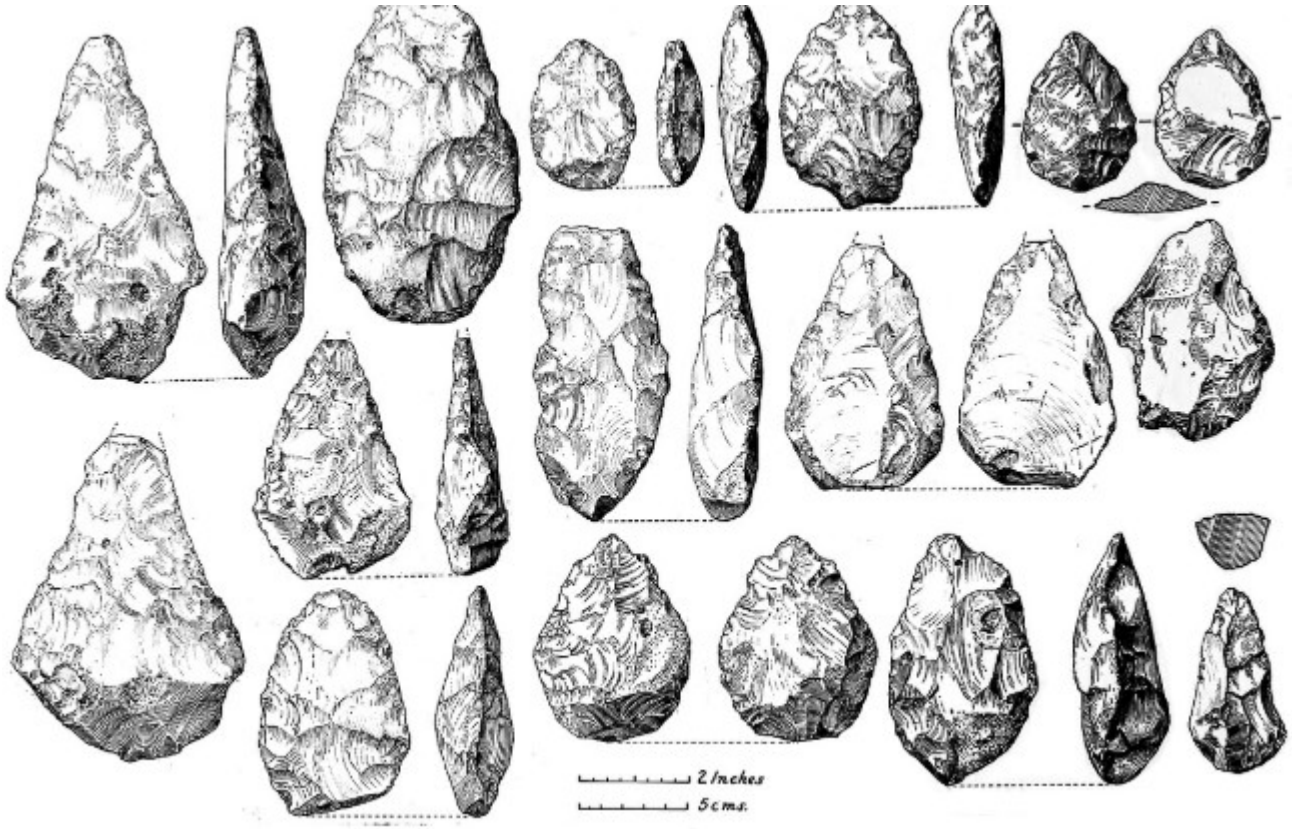
Homo erectus Culture



- Modified environment to suit own purposes
- Home bases:
  - Built shelters to protect from the elements
  - *Terra Amata:* site dating to 400 000 years ago
    - Wood huts
    - Evidence of fire use
    - Tools from stone and bone
    - Remnants of animal bone from hunting - also fish bones – fishing?
- Use of fire?
  - Evidence that *H.erectus* used fire:
    - Kept predators away
    - Warmth and light
    - Used during hunting to drive animals
    - Cooking of food
- Tools
  - Flaked around all edges, teardrop shaped “hand axes”
  - Referred to as *Acheulean tools* (from where they were first discovered)

Homo erectus Culture

- Food:
  - Hunter- gatherer society
    - Most food from gathering, supplemented by hunting
  - Skilful, strategic hunters
    - Evidence of planning – planned baboon slaughter miles away from base
    - Drove elephants into swamps using fire, then butchered them
    - Sophisticated, systematic butchering using tools
  - Women used complex techniques and collaboration to gather food.
    - Balanced with child rearing
- Social organisation
  - A result of hunting/gathering/use of tools and fire
  - Complex society, shared care
  - Relatively complex social language may have developed.



Acheulian tools – Homo erectus

- Rough tear-drop shape
- Not a lot of variation
- General rather than specific use
- Finer flaking than Oldowan
- Bifacial – thin axe-like edges with two broader faces.

Neanderthal Culture

- Further cultural advances – well able to manipulate environment
- Tool making
  - Production of stone flakes which could then be trimmed to form various tools.
  - **“Mousterian Industry”** – advance over hand axes
  - Allowed clothing to be made



Figure 20.7 Neanderthal tools

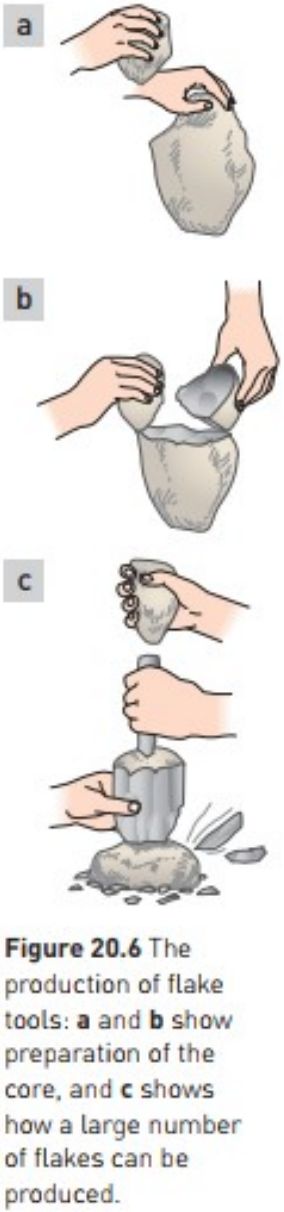
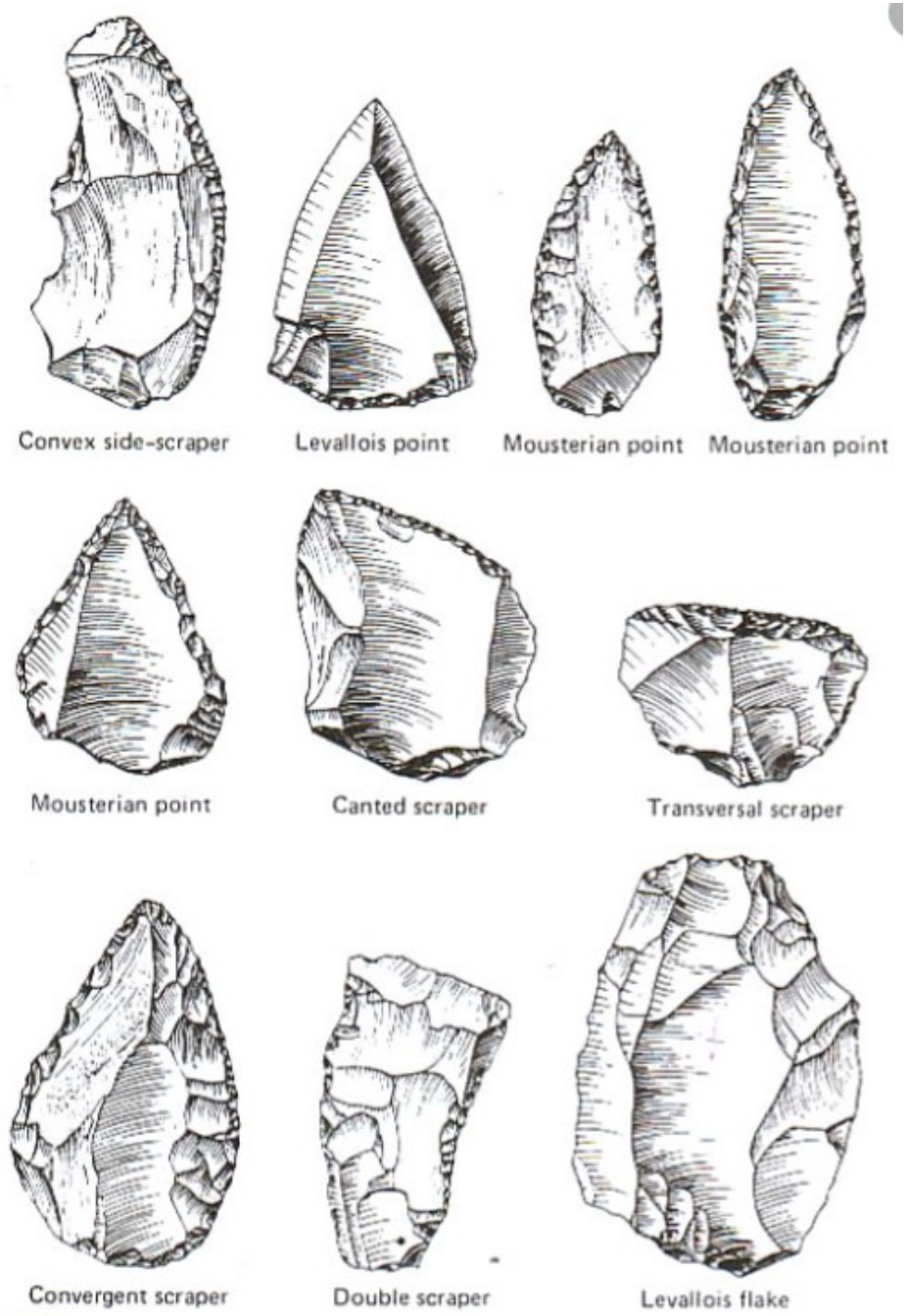


Figure 20.6 The production of flake tools: **a** and **b** show preparation of the core, and **c** shows how a large number of flakes can be produced.

Mousterian Tools – Neanderthal Culture

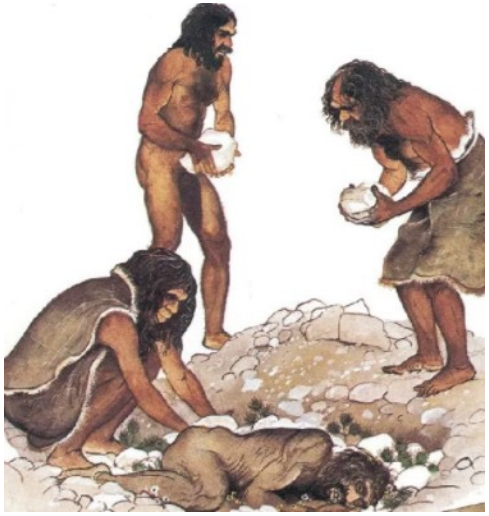
- Variety of tool shapes to suit different purposes eg scraping, stabbing
- Relatively fine flaking to edges for sharpness



Neanderthal Culture

- Burial of dead
  - With symbolic objects eg goat horns, flowers
  - Evidence of spiritual/religious belief?

- Shared care and responsibility
  - Likely that Neanderthals took care of the disabled and elderly
  - Developed social system for sharing of food and other resources



Transition to Modern Humans

- 50 000 years ago – new technologies
  - Finer blades
  - Projectile weapons
  - Sewn clothing
  - Better shelters
- Blade tools:
  - Sharp, parallel sides – like a knife blade
- Sophisticated hunting techniques
  - Meat, skins, oil for fire/lamps
  - Bones and ivory used to make tools

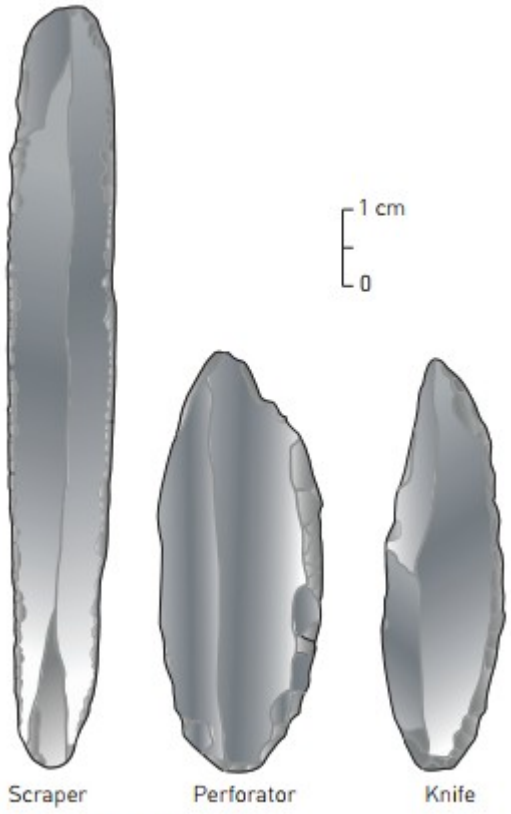
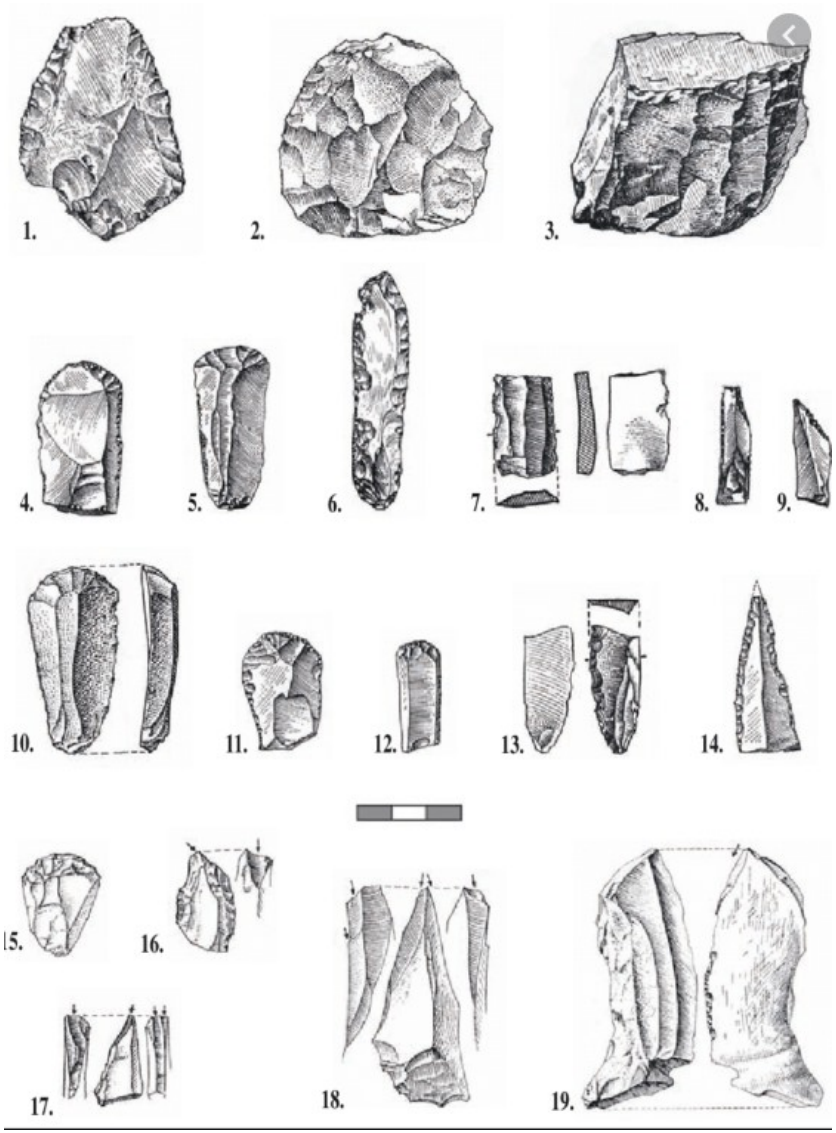


Figure 20.8 Blade tools

Transition to Modern Humans

- 3 main cultures in fossil record for *Homo sapiens*
  - Aurignacian (40 000 – 28 000 bce)
  - Solutrean (21 000 – 17 000 bce)
  - Magdelanian (17 000 – 12 000 bce)

Aurignacian Tools



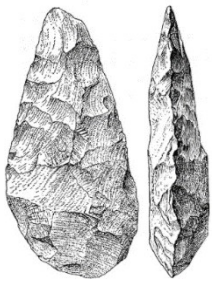
40 000 – 28 000 bce

- Finer workmanship
- Wide variety of tools for different purposes

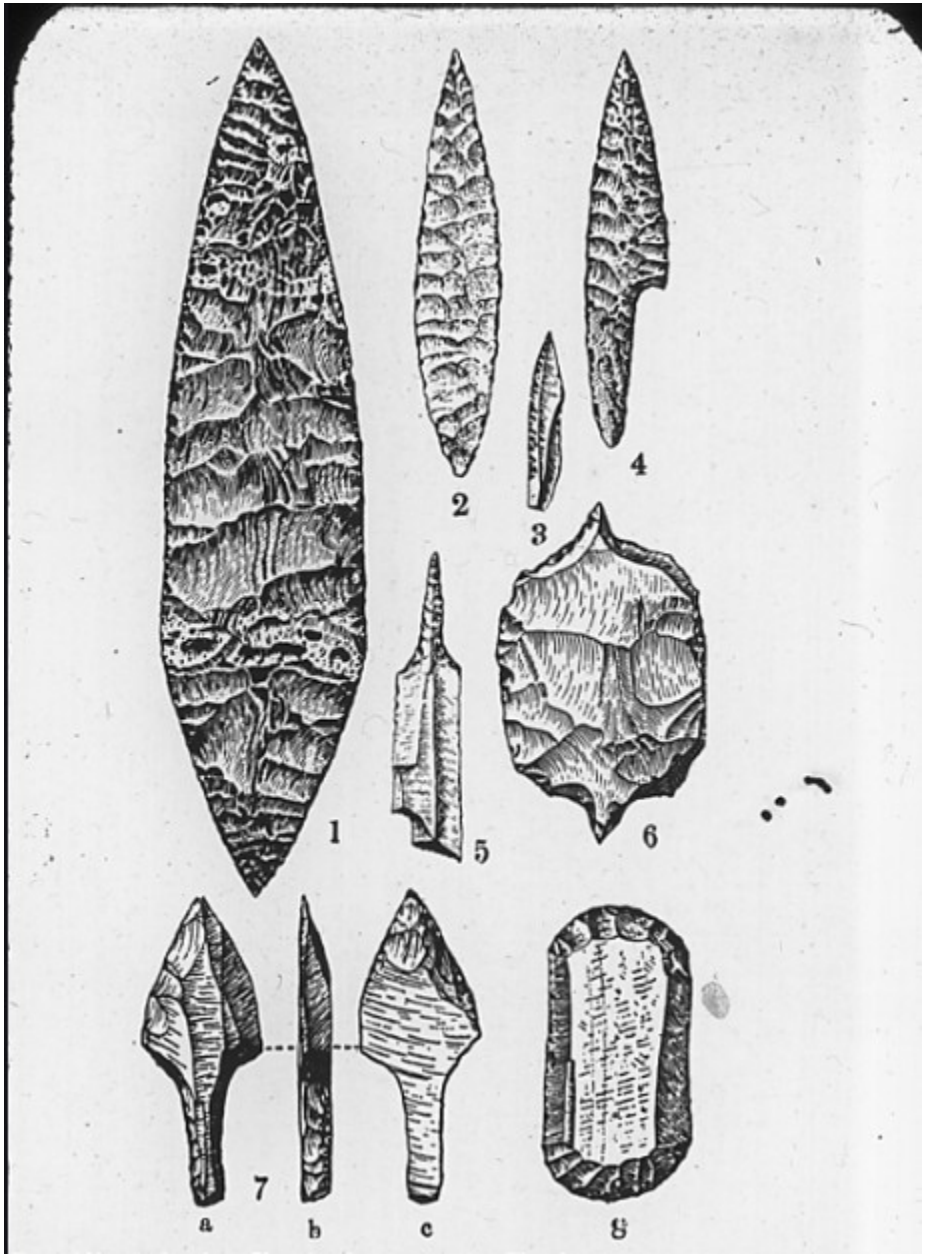
Solutrean Tools

- 21 000 – 17 000 bce
  - Variety of tools were made for different purposes
  - Complex, fine blades, pressure flaking
  - Symbolic blades were made in some instances.

Note: comparing Solutrean to Acheulian (Homo erectus) tools - solutrean tools have finer flaking and are much smaller and more elongated than the H. erectus tools



Acheulian hand axe



Transition to Modern Humans

- Magdelanian
  - Bone and antler tools predominated
    - Barbed spears
    - Spear throwers
    - needles
  - Tool for making other tools – a stone “burin” chisel used for carving bone and antler
  - Some clay modelling
  - Art works
    - Portable – carved figurines
    - Mural art on cave walls
    - Use of colour and shading

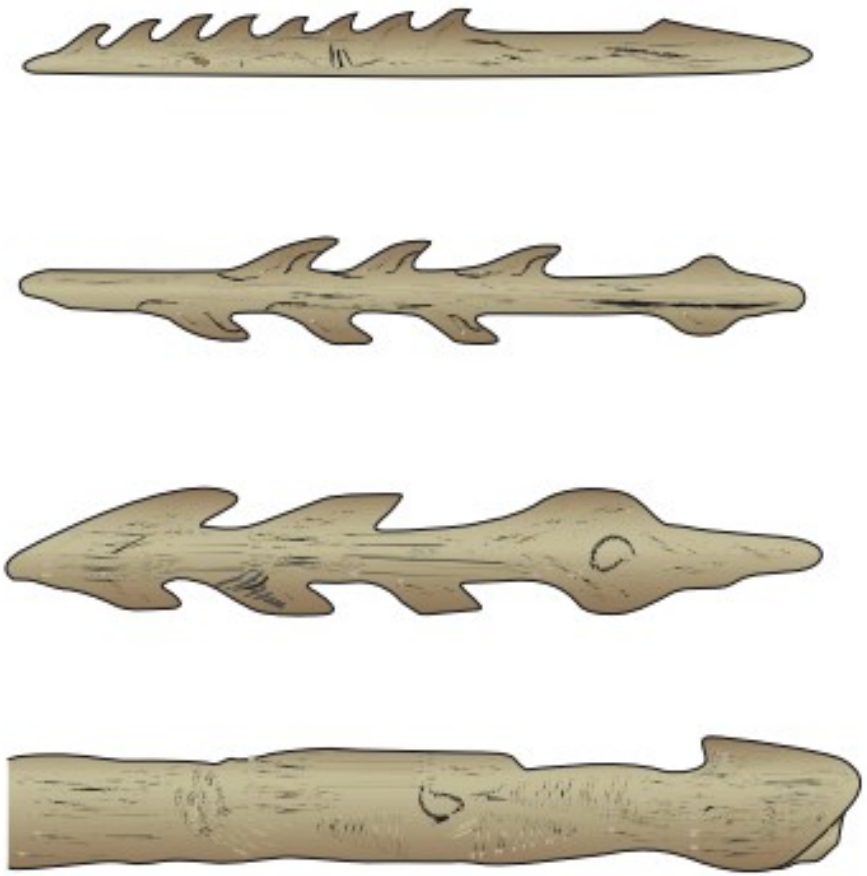


Figure 20.10 Magdalenian barbed points and spear thrower (bottom) made of bone or antler

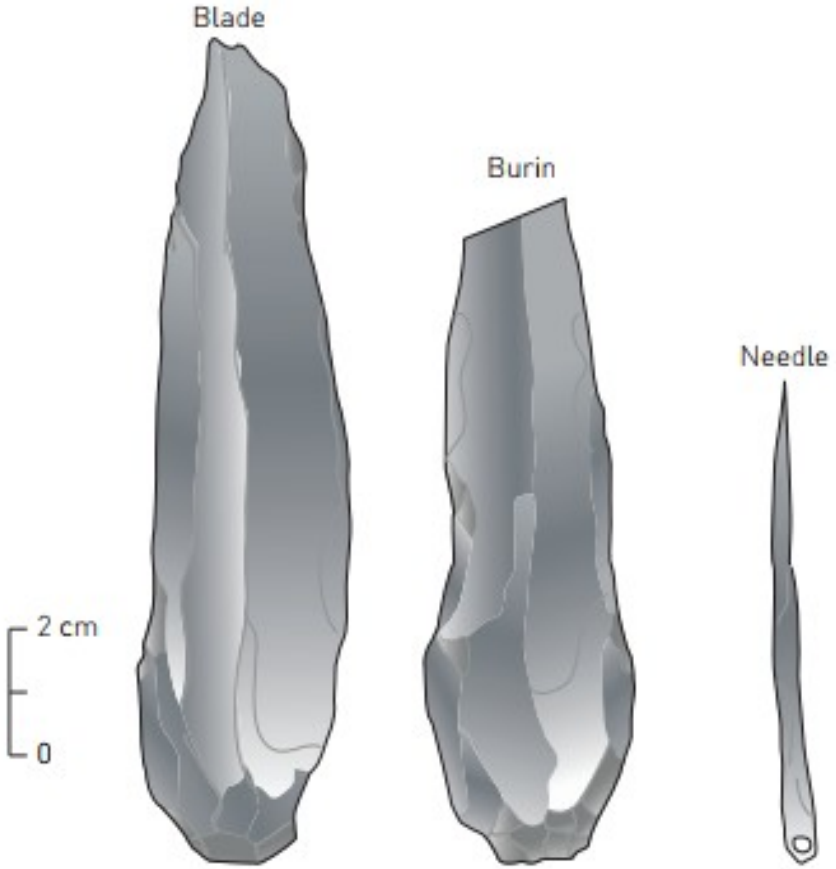


Figure 20.11 A burin was made from a basic blade. It was used to cut into antler and bone to make other tools, such as the needle.



Figure 20.12 Cro-Magnon carving: **a** a Venus statuette carved from sandstone and **b** a horse carved from ivory



Philippe Psaila/Science Photo Library

Figure 20.13 Drawings representative of Cro-Magnon artwork

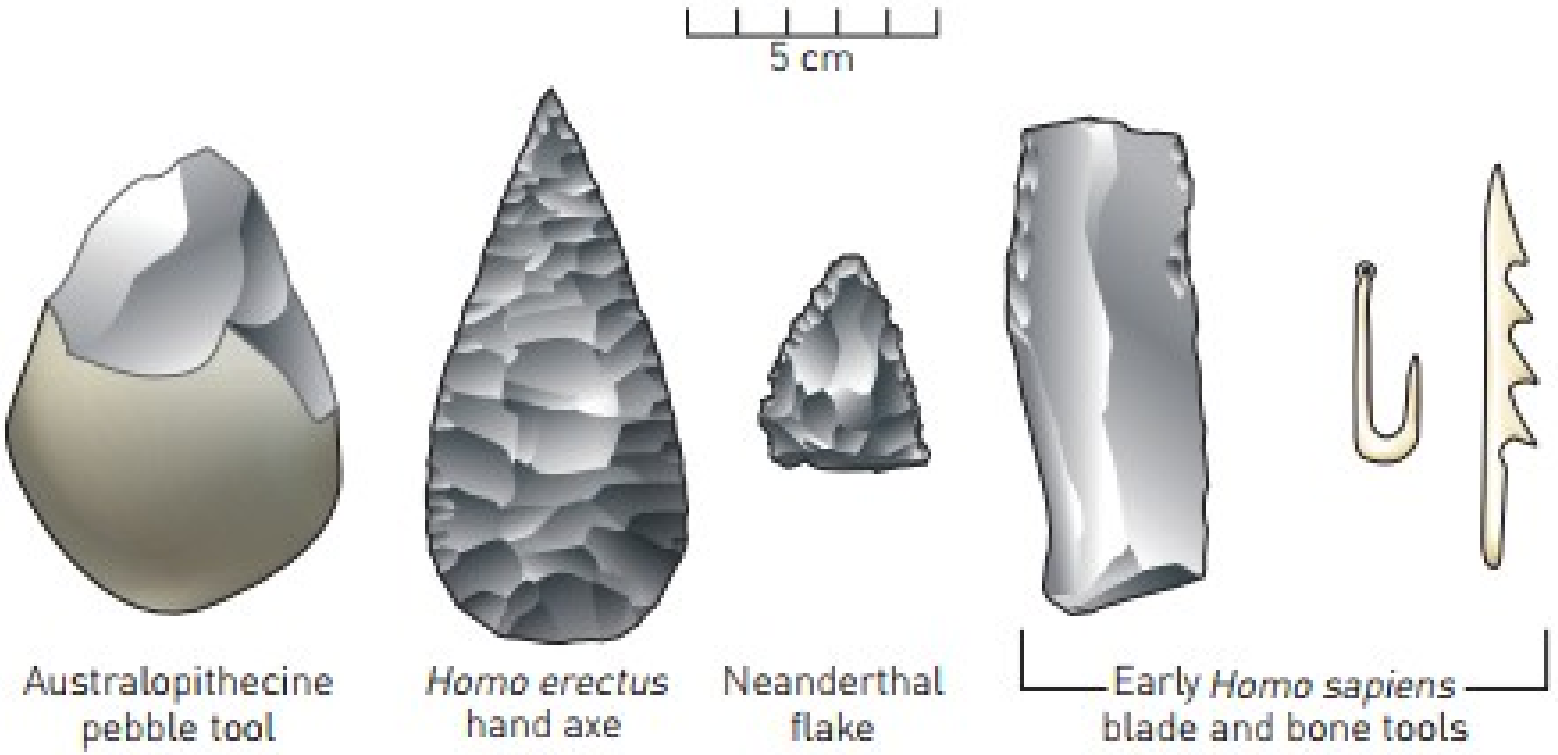


Figure 20.14 An example of cultural evolution – increasing sophistication of simple tools from australopithecines to early *Homo sapiens*

The beginning of agriculture

- Big step in human history
- Change from nomadic to village lifestyle
- “Neolithic revolution”
- Domestication of plant and animal species
- Likely to have begun in South-West Asia, around Palestine
  - “Fertile Crescent”
  - 12 500 years ago
  - Gathered wild grasses, which hybridised
  - Fertile hybrid: “Emmer”
  - Later hybrid; large grains, good for making flour, but could no longer by scattered by wind: had to be sown – agriculture!

The beginning of agriculture

- Domestication of crops occurred independently at different sites around the world:
  - China – rice, millet, soybeans, yams
  - Central America – maize (corn), potato, sweet potato, bean, peanut, pumpkin, tomato
  - New World; more plants domesticated. Old World – more animals domesticated
- 11 000 years ago: animals domesticated
  - Sheep – south west asia
  - Pigs, cattle, goats, and other animals, over a longer period of time
- Farming supplemented by fishing and hunting
- Surplus grain could be stored
- More people could be supported on less land
- Settled life allowed development of new technologies



Figure 20.17 Wild wheat

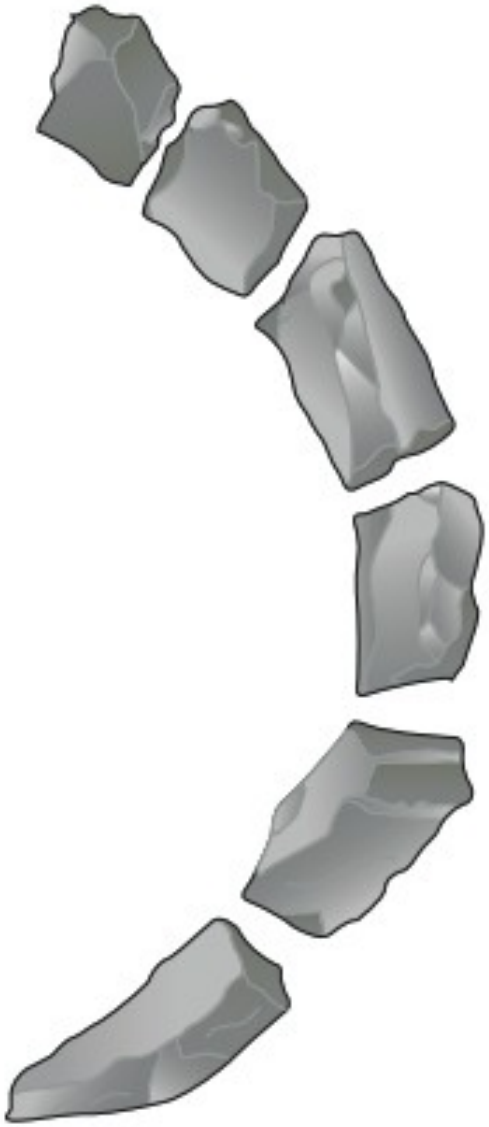


Figure 20.18 Pieces of flint used as blades for a sickle to harvest wheat



Figure 20.19 The region of early settlement around the Persian Gulf and eastern Mediterranean, often referred to as the "fertile crescent"

Modern Humans

- We are able to manipulate our environment to a high level
  - Travel
  - Medical technology – life expectancy
  - Complex society

- Important to understand human origins
  - Helps understand modern humans
  - Lifestyle diseases often because our cultural evolution has outstripped our biological evolution
- Still challenges to survival, some of our own making
  - Climate change an immediately pressing issue
  - May be able to evolve culturally so we are able to slow down or reverse the change we are making.